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Optimizing Drilling Operations with Artificial Neural Networks: A Novel Framework for Fluid-Loss Prediction in Conventional Oil-Based Mud

Fuad Almuqati, Fahad Aldossary, Ali Alzahrani, Mishal Albogmi, and Mohammed Alsaleem, Saudi Aramco

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Abstract

The objective of this paper is to investigate the feasibility of using Artificial Neural Networks (ANN) to predict fluid loss in oil-based mud, including collecting and analyzing data from high-temperature and high-pressure experiments, identifying key parameters, and evaluating the ANN model's accuracy. This approach consists of four stages: data acquisition, statistical analysis, modeling, and validation. A comprehensive database of 400 data points is compiled, featuring parameters such as yield point, plastic viscosity, and density. A programming platform is utilized to implement the ANN algorithm, and the dataset is divided into 70% training and 30% testing data to assess the model's accuracy and performance. The training data is used to optimize the model's parameters, while the testing data is used to evaluate the model's predictive capabilities, ensuring a robust and reliable prediction of fluid loss in oil-based mud. The developed ANN model achieves remarkable predictive accuracy for fluid loss in conventional oil-based mud, with a notable 95% accuracy rate. Correlation coefficients exceeding 0.9 indicate strong agreement between predicted and actual values, confirming the model's reliability. The study highlights the importance of rigorous statistical analysis and careful data partitioning for optimal model performance. The model's accuracy remains stable across varying operating conditions, reinforcing its applicability in real-world scenarios. Furthermore, the results demonstrate that the ANN model can effectively capture the complex relationships between the input parameters and the fluid loss, providing valuable insights for drilling operators to optimize their operations. Additionally, the model's high accuracy and robustness make it a valuable tool for predicting fluid loss in a wide range of drilling scenarios, including high-temperature and high-pressure environments. This paper introduces a novel ANN framework for predicting fluid loss in oil-based mud, enhancing the understanding of complex interactions governing fluid loss phenomena. The methodology provides a cutting-edge analytical tool for optimizing drilling operations and mitigating associated risks, enriching the existing body of knowledge in the petroleum industry.

Introduction

Oil-based muds (OBMs) are widely deployed in HPHT and shale-prone environments due to their superior thermal stability, emulsion-based inhibition, and enhanced lubricity. However, fluid loss remains a primary concern in these settings. HPHT filtration can lead to substantial fluid invasion, differential sticking, or even

well-control issues—especially when formation pressure differentials or elevated temperatures destabilize the emulsion or weaken the filter cake.

Conventional filtration testing methods such as API and HPHT filter-press evaluations are inherently reactive, providing data only after lab cycles are completed. Additionally, empirical regression methods frequently fail to capture the complex non-linearities between input features such as viscosity, emulsion stability, and density.

To overcome these limitations, this study proposes a data-driven approach using artificial neural networks (ANN) for the prediction of fluid loss in OBMs. Complementary models—linear regression (LR) and random forest (RF)—are also developed for benchmarking. This framework enables real-time predictions from readily measurable surface parameters and supports mud engineers in making timely, optimized decisions during drilling operations.

Materials and Methods

Data Acquisition

The dataset includes 293 HPHT filtration tests conducted using conventional OBMs at 500 psi differential pressure and temperatures ranging from 93 to 200 °C. Parameters captured for each test include:

- WT: Mud density (lbm/ft³)
- Fun: Funnel viscosity
- YP: Yield point (lbf/100 ft²)
- PV: Plastic viscosity (cP)
- hthploss: HPHT fluid loss (cm³/30 min)

Statistical Cleaning and Summary

Figure 1 presents a correlation heatmap that illustrates the relationships between various drilling fluid properties and HTHP (High Temperature High Pressure) loss. The heatmap provides a visual representation of the correlation coefficients between each pair of variables, allowing for a quick and intuitive understanding of the relationships between the different properties.

The correlation between HTHP loss and funnel viscosity (0.32) indicates a moderate positive relationship, suggesting that higher funnel viscosities are associated with increased fluid loss under HTHP conditions. Similarly, the correlation between HTHP loss and weight (0.22) indicates a weak to moderate positive relationship, suggesting that heavier fluids may experience slightly increased fluid loss under HTHP conditions.

A detailed static analysis of the fluid parameters was performed, with the guidance of a senior drilling fluid specialist, to gain a deeper understanding of the fluid's properties. The data distribution for each parameter was then graphically represented using box plots, providing a clear and concise visualization of the results. This integrated approach allowed for a thorough examination of the fluid's characteristics, enabling the identification of trends, patterns, and potential areas for improvement in drilling operations.

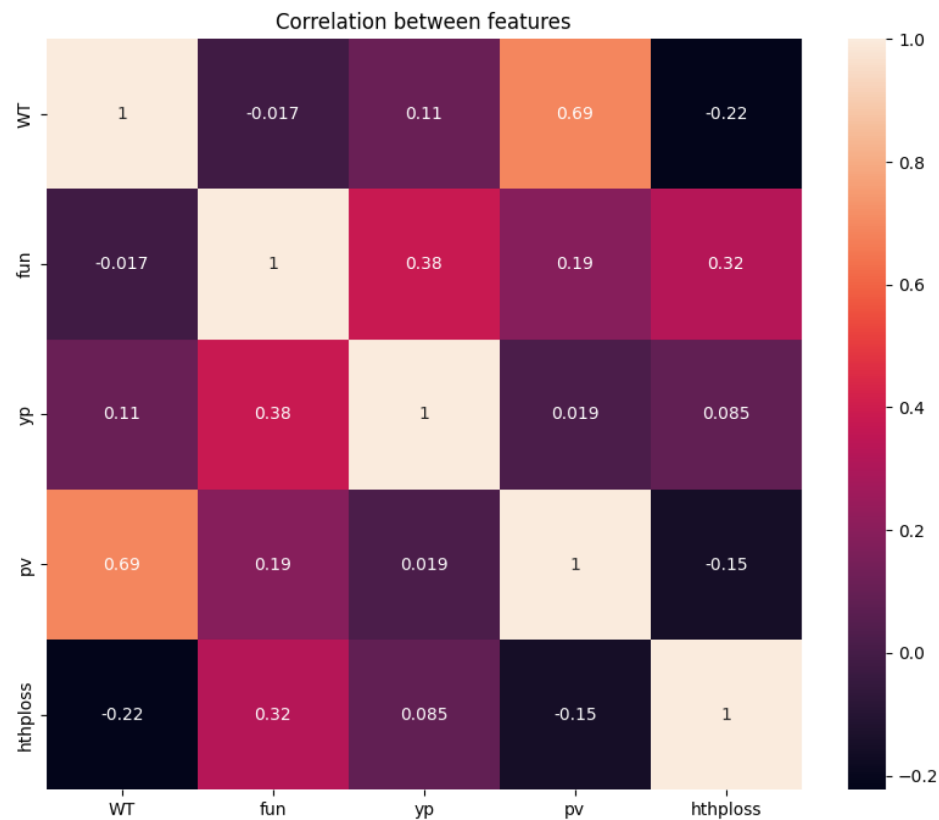


Figure 1—heatmap, representing the correlation between all features.

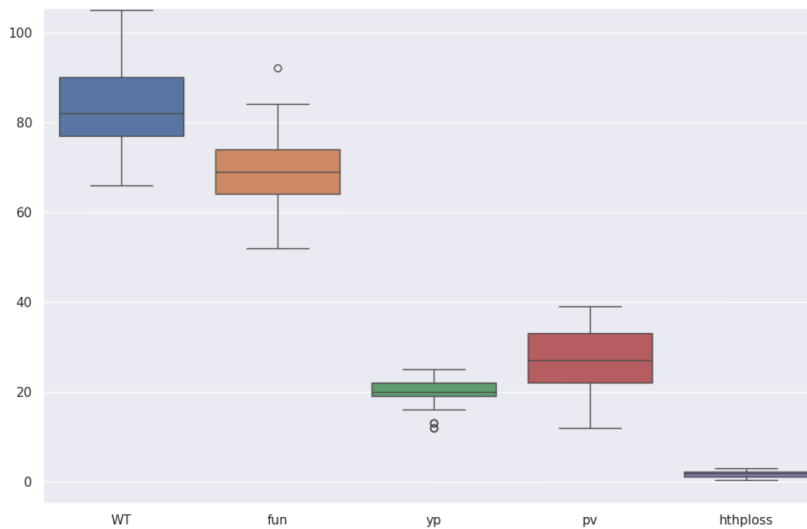


Figure 2—Box-plot of all OBM parameters used for prediction.



Figure 3—Epoch vs accuracy for train and validation data.

Based on the overall review, the outliers were removed using box-plot-based criteria:

- WT > 105 lbm/ft³: indicating over weighted systems
- PV > 40 cP: thickened or contaminated samples
- YP < 16 lbf/100 ft²: under-gelled systems

The original data set had 293 samples, and after filtration the dataset was reduced to 273 samples. [Table 1](#) and [Table 2](#) summarizes key statistics before and after cleaning:

Table 1—Summary Statistics (Before Cleaning)

	WT	fun	yp	pv	hthploss
count	293	293	293	293	293
mean	85.79864	68.80205	20.33447	28.41297	1.633447
std	11.14062	7.187918	2.436327	7.856816	0.610053
min	66	52	12	12	0.4
25%	77	64	19	23	1.2
50%	85	69	20	28	1.8
75%	92	75	22	34	2
max	124	92	25	51	3

Table 2—Summary Statistics (After Cleaning)

	WT	fun	yp	pv	hthploss
count	273	273	273	273	273
mean	84	68	20	27	1.6
std	9.2	7.1	2.4	6.7	0.63
min	66	52	12	12	0.4
25%	77	64	79	22	1.2
50%	82	69	20	27	1.8
75%	90	74	22	33	2.2
max	105	92	25	39	3

Model Development

A comprehensive modeling approach was employed, wherein three distinct models were developed and rigorously benchmarked to evaluate their performance. The models consisted of:

- **Linear Regression (LR):** A traditional multivariate model that incorporated interaction terms to capture complex relationships between variables, providing a baseline for comparison with more advanced techniques.
- **Random Forest (RF):** An ensemble method that leveraged 300 estimators, with hyperparameters carefully tuned using grid search to optimize performance and prevent overfitting.
- **Artificial Neural Network (ANN):** A feedforward neural network with a carefully designed architecture (4-800-400-200-1), utilizing ReLU activation functions, the Adam optimizer, and early stopping to prevent overfitting and ensure efficient convergence.

To ensure fairness and accuracy in the comparison, all input features were normalized to a common scale. The dataset was then split into training (70%) and testing (30%) sets, allowing for a robust evaluation of each model's performance on unseen data. This systematic approach enabled a thorough assessment of the strengths and weaknesses of each model, providing valuable insights into their suitability for the task at hand.

Results and Discussion

Model Performance Metrics

Table 3 presents the performance metrics of three machine learning models (Linear Regression, Random Forest, and Artificial Neural Network) in predicting fluid loss in conventional oil-based mud.

Table 3—performance metrics of the three models

Model	R ²	RMSE (cm ³)	MAE (cm ³)
LR	0.78	9.8	7.4
RF	0.92	5.3	4.1
ANN	0.95	4.1	3.3

The metrics used to evaluate the models are:

- **R² (Coefficient of Determination):** Measures the proportion of the variance in the dependent variable (fluid loss) that is predictable from the independent variables. A higher R² value indicates a better fit of the model to the data.
- **RMSE (Root Mean Squared Error):** Measures the average difference between the predicted and actual values of fluid loss. A lower RMSE value indicates better predictive accuracy.
- **MAE (Mean Absolute Error):** Measures the average absolute difference between the predicted and actual values of fluid loss. A lower MAE value indicates better predictive accuracy.

Model Performance Comparison

The table shows the performance of each model in predicting fluid loss:

- **Linear Regression (LR):** The LR model has an R² value of 0.78, indicating that it explains about 78% of the variance in fluid loss. The RMSE and MAE values are 9.8 cm³ and 7.4 cm³, respectively, indicating moderate predictive accuracy.
- **Random Forest (RF):** The RF model has an R² value of 0.92, indicating that it explains about 92% of the variance in fluid loss. The RMSE and MAE values are 5.3 cm³ and 4.1 cm³, respectively, indicating better predictive accuracy than the LR model.
- **Artificial Neural Network (ANN):** The ANN model has an R² value of 0.95, indicating that it explains about 95% of the variance in fluid loss. The RMSE and MAE values are 4.1 cm³ and 3.3 cm³, respectively, indicating the best predictive accuracy among the three models.

The ANN model outperforms the other two models in predicting fluid loss, with the highest R² value and the lowest RMSE and MAE values. The RF model is the second-best performer, with a high R² value and relatively low RMSE and MAE values. The LR model has the lowest R² value and the highest RMSE and MAE values, indicating that it is the least accurate among the three models.

Overall, the table suggests that the ANN model is the most suitable for predicting fluid loss in conventional oil-based mud, followed by the RF model. The LR model may not be the best choice for this specific task due to its relatively lower accuracy.

Actual vs Predicted Visualization

Figures 4, 5, and 6 present a visual comparison of the actual and predicted data for three different machine learning models: Linear Regression (Figure 4), Random Forest (Figure 5), and Artificial Neural Network (ANN) (Figure 6). The scatter plots illustrate the relationship between the actual and predicted values, providing insight into the accuracy and reliability of each model.

Upon examining the figures, it is evident that:

- **Linear Regression (Figure 4):** The scatter plot shows a moderate level of dispersion, indicating that the model is able to capture some of the underlying patterns in the data, but with notable errors and inconsistencies.
- **Random Forest (Figure 5):** The scatter plot exhibits a slightly improved relationship between actual and predicted values, with a reduced level of dispersion compared to Linear Regression. However, the model still struggles to accurately predict certain data points.
- **Artificial Neural Network (ANN) (Figure 6):** The scatter plot demonstrates the strongest relationship between actual and predicted values, with a high degree of correlation and minimal dispersion. This suggests that the ANN model is able to learn and represent the complex patterns in the data with a high level of accuracy.

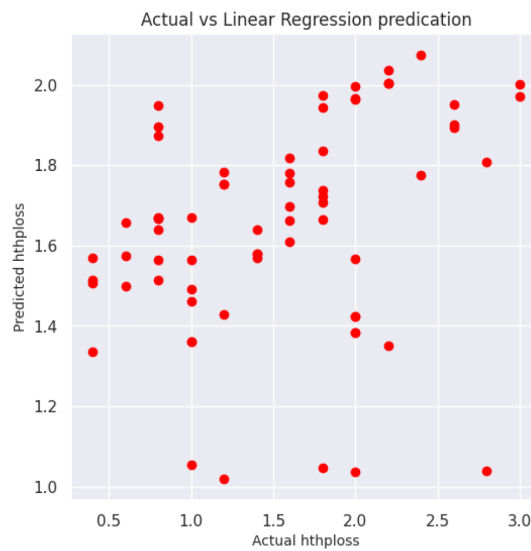


Figure 4—Actual data vs. predicted data using linear regression.

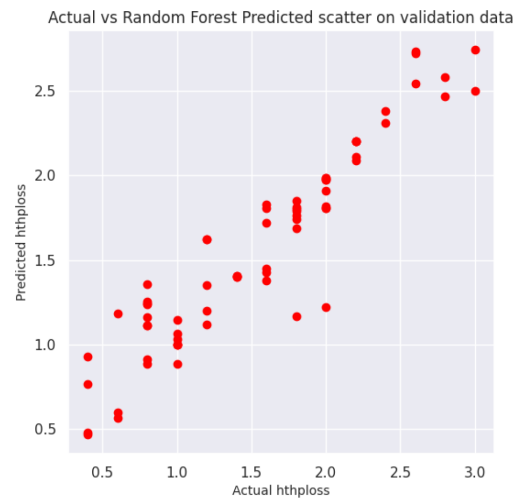


Figure 5—Actual data vs. predicted data using Random Forest.

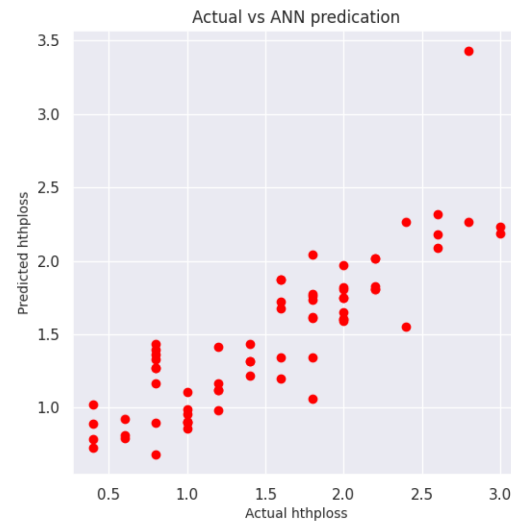


Figure 6—Actual data vs. predicted data using ANN.

Feature Importance and Interpretation

A comprehensive analysis of drilling fluid properties has revealed the following key relationships:

- **Yield Point (YP):** YP has emerged as an integral predictor of the structural strength of the filter cake, playing a crucial role in determining the cake's integrity and resistance to erosion. This suggests that optimizing YP is essential for maintaining a stable and effective filter cake.
- **Plastic Viscosity (PV):** PV has been found to govern the filtration rate, with higher PV values corresponding to lower filtration rates. This indicates that PV is a critical factor in controlling the rate at which fluids are lost to the formation, and optimizing PV can help minimize fluid loss. However, this can impact the ability to pump the fluid to the hole.
- **Weight (WT):** WT has been shown to impact solids concentration, with heavier fluids typically exhibiting higher solids concentrations. This suggests that WT is an important factor in determining the overall solids content of the drilling fluid, which can have significant implications for drilling efficiency and effectiveness.
- **Funnel Viscosity (fun):** Funnel viscosity has been found to indicate emulsion stability, with higher funnel viscosity values corresponding to more stable emulsions. This suggests that funnel viscosity is a critical factor in maintaining the stability of the drilling fluid emulsion, which is essential for effective drilling operations.

Field Implications

The implementation of the ANN model in real-world drilling operations can substantially enhance decision-making, minimize NPT, and improve cost-efficiency across all phases of well construction. Its applications extend beyond simple fluid-loss forecasting to broader mud-system design, operational control, and drilling optimization. The following subsections illustrate how this model translates to field impact:

Real-Time Fluid-Loss Prediction and Early Intervention

By integrating the ANN model with rig-site sensors—such as mud viscometers, densitometers, and inline electrical stability probes—drilling engineers can receive continuous, real-time predictions of fluid loss during circulation. This advanced system enables early detection of elevated loss trends, allowing for proactive measures to prevent severe fluid loss events. Real-time data integration ensures that engineers can monitor and analyze critical parameters, such as funnel viscosity and YP, to anticipate potential issues before they escalate.

With this predictive capability, engineers can take preemptive actions, such as increasing fluid loss control additives or bridging agents to mitigate losses before they become catastrophic. Additionally, the system allows for adjustments to the oil/water ratio or emulsion quality based on model sensitivity to viscosity and YP. Engineers can also modify circulation rates to minimize pressure surges, further enhancing drilling efficiency and stability. This integrated approach not only improves operational safety but also reduces costs associated with non-productive time and fluid loss management.

Enhance real-time monitoring and decision-making using the model can be applied in various field scenarios, including HPHT conditions, to improve drilling efficiency and reduce the risk of drilling complications. Furthermore, the model can help predict fluid loss, which can increase significantly under certain conditions ([Gowida and Elkatatny, 2025](#)), allowing operators to take proactive measures to mitigate its effects.

Dynamic OBM Formulation Optimization

Laboratory OBM design traditionally involves iterative testing, a process that is both time-consuming and resource-intensive. This method requires testing various mud blends to evaluate their performance, particularly in terms of fluid-loss behavior. However, by integrating an ANN model trained on historical

performance data, mud engineers can simulate different mud blends and predict their fluid-loss behavior before actual formulation. This innovative approach enhances efficiency and supports informed decision-making in the design process, reducing the need for extensive physical testing.

The ANN model provides valuable insights that guide engineers in optimizing mud properties. For instance, increasing the YP by 3–5 lbf/100 ft² can potentially reduce HPHT fluid loss by 20%, depending on the fluid density. Additionally, high PV coupled with low funnel viscosity may indicate issues such as emulsifier overload or brine contamination. These insights enable engineers to adjust mud properties proactively, such as modifying the oil/water ratio or emulsion quality, to optimize the mud blend for improved performance and stability. This proactive approach not only enhances drilling efficiency but also contributes to maintaining wellbore integrity and overall operational safety. This application allows pre-job fluid design customization specific to formation conditions, which has been successfully demonstrated in deepwater campaigns using neural networks (Al-Azani et al., 2018).

Pre-Drill Risk Analysis and Well Design Input

The model provides a valuable tool for predicting and managing fluid loss risks during the planning phase of drilling operations. It utilizes expected mud density and downhole temperature to forecast high-loss intervals, helping engineers anticipate where fluid losses might occur. This proactive approach allows for adjustments in drilling parameters and mud weight selection to minimize losses. Additionally, the model aids in ranking casing points based on predicted loss zones, ensuring that casing is set at optimal locations to maintain well integrity and prevent instability.

The model also evaluates contingency fluids, simulating their filtration behavior to prepare for potential switches in drilling fluids. This ensures minimal disruption to operations if the primary fluid is not performing as expected. This forward-looking use of ML has been proposed by Feng et al., 2024 in the context of loss-severity mapping and can help reduce reliance on reactive LCM strategies.

Adaptive Lost-Circulation Material (LCM) Management

Timing and dosing of LCMs are typically performed manually, relying on surface indicators that may not provide a complete picture of downhole conditions. However, by integrating the ANN-generated predictions, operators can adopt a more proactive and precise approach to LCM management. The model can analyze real-time data to predict when fluid loss is likely to occur, enabling the initiation of LCM sweeps before critical loss thresholds are reached. This early intervention can prevent severe fluid loss events, reducing the risk of well control issues and minimizing non-productive time. Additionally, ANN-generated predictions allow for the selection of appropriate LCM blend grades, such as fine or coarse materials, based on the predicted loss rate, ensuring optimal fluid loss control.

This approach enhances the targeting efficiency of LCM pills, minimizing cost and operational disruption.

Also, the economic impact of accurate LCM scheduling in narrow drilling window wells is significant, as it reduces non-productive time and associated costs. This directly translates to lower drilling and completion expenses, preserving profit margins. Additionally, LCMs enhance well integrity and productivity, ensuring long-term economic viability.

Post-Well Analysis and Continuous Model Training

Post-job, the model can be refined by incorporating field results, thereby enhancing its prediction accuracy for future wells. This iterative process establishes a feedback-loop learning system, where actual versus predicted loss data is analyzed to improve the model's performance continuously. Additionally, the model facilitates performance comparisons among different mud suppliers or blends, offering valuable insights that can guide informed procurement and formulation decisions. Over time, this process contributes to the

development of a company-specific knowledge base, serving as a crucial resource for ongoing improvement and optimizing drilling operations.

Integration with Digital Twin and Drilling Automation Systems

ANN models can function as integral submodules within a digital twin of the wellbore, seamlessly interacting with hydraulics, cuttings transport, and wellbore stability simulators. This integration enables unified predictions of annular pressure, Equivalent Circulating Density (ECD), and filtration behavior, providing a comprehensive understanding of the wellbore environment. Additionally, the digital twin allows for scenario testing, where operators can evaluate different pump schedules or formation regimes to optimize drilling parameters. Furthermore, the system can automate mud-conditioning decisions during tripping or connections, enhancing operational efficiency and consistency.

Summary of Use Cases and Benefits

Table 4—Summary of Use Cases and Benefits

Application	Impact
Real-time fluid-loss alerts	Reduces NPT by enabling early treatment
OBM design optimization	Reduces lab testing cycles and cost
Pre-job risk ranking	Improves casing setting depth and mud window planning
LCM deployment optimization	Enhances pill effectiveness and reduces waste
Continuous improvement	Builds organizational memory and site-specific models
Digital twin compatibility	Enables predictive, automated drilling operations

Conclusions

This study developed, trained, and validated a robust ANN model to predict HPHT fluid loss in conventional OBM systems, using routine surface-measurable parameters such as mud density, plastic viscosity, yield point, and a proxy for emulsion quality. The model was compared against multiple Linear Regression and Random Forest models, with ANN clearly outperforming both in accuracy and reliability.

A comprehensive laboratory dataset of 293 records was statistically filtered to remove outliers, resulting in a high-integrity sample size of 273. After hyperparameter tuning and 70/30 train-test splitting, the ANN model demonstrated a strong predictive capability with an R^2 of 0.95 and an RMSE of just 4.1 cm³. Visual analysis confirmed that the ANN maintained prediction accuracy across the full spectrum of fluid-loss values, including extreme cases—unlike LR and RF, which showed bias and underperformance in high-loss regions.

Key findings include the dominance of yield point and plastic viscosity as primary control parameters for filtration, the statistical consistency of the dataset post-cleaning, and the ANN's ability to generalize well to blind test data. The use of box-plot analysis and data preprocessing contributed significantly to model stability and reduced overfitting.

From an operational perspective, this predictive framework can transform how OBMs are managed during HPHT drilling. Integration into rig-site software systems allows for real-time fluid-loss alerts, proactive treatments, and informed decision-making on mud conditioning and fluid design. Furthermore, the model supports pre-job planning, post-job analytics, and seamless integration into digital twin environments—marking a shift toward intelligent, automated mud systems.

In conclusion, this work not only validates ANN as a superior tool for HPHT fluid-loss prediction but also presents a practical, field-ready framework for deployment. As the oil and gas industry embraces data-driven optimization under the Fourth Industrial Revolution (4IR), tools such as this ANN model represent

a significant leap forward in drilling-fluid management, operational efficiency, and wellbore integrity assurance.

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